

Assignment 1: Math1014

September 7, 2026

1 Linear Algebra

1.1 Question 1

we are given the vector space

$$V_N = \text{span} \{ \cos(2\pi kt), \sin(2\pi kt) \mid 0 \leq k \leq N \}.$$

(a)

two functions are linearly independent if the only scalars a, b satisfying

$$a \cos(2\pi t) + b \sin(2\pi t) = 0 \quad \text{for all } t \in [0, 1]$$

are $a = b = 0$. if we can find even one nontrivial choice of t that forces $a = 0$, and another that forces $b = 0$, we're done.

- at $t = 0$,

$$a \cos(0) + b \sin(0) = a(1) + b(0) = a = 0$$

so a must be 0

- at $t = \frac{1}{4}$

$$a \cos\left(\frac{\pi}{2}\right) + b \sin\left(\frac{\pi}{2}\right) = a(0) + b(1) = b = 0$$

so b must be 0 too

since the equation forces $a = b = 0$, the only linear combination equal to the zero function is the trivial one. hence $\cos(2\pi t)$ and $\sin(2\pi t)$ are **linearly independent** in V_1 .

(b)

we need the dimension of V_1

the definition says

$$V_N = \text{span} \{ \cos(2\pi kt), \sin(2\pi kt) \mid 0 \leq k \leq N \}$$

if $N = 1$, then k can be either 0 or 1.

so we get

$$V_1 = \text{span} \{ \cos(0), \sin(0), \cos(2\pi t), \sin(2\pi t) \}$$

now, $\cos(0) = 1$ and $\sin(0) = 0$
so the span becomes:

$$V_1 = \text{span} \{ 1, \cos(2\pi t), \sin(2\pi t) \}$$

the zero function doesn't add anything to the span, so we can ignore it
we therefore have three functions:

$$1, \cos(2\pi t), \sin(2\pi t)$$

these are linearly independent, so they form a basis for V_1 .
therefore, $\boxed{\dim(V_1) = 3}$

(c)

finding the dimension of V_2 ,
when $N = 2$, the possible values of k are 0, 1, 2
therefore,

$$V_2 = \text{span} \{ \cos(0), \sin(0), \cos(2\pi t), \sin(2\pi t), \cos(4\pi t), \sin(4\pi t) \}$$

again, $\cos(0) = 1$ and $\sin(0) = 0$, so (leaving out the 0 terms),

$$V_2 = \text{span} \{ 1, \cos(2\pi t), \sin(2\pi t), \cos(4\pi t), \sin(4\pi t) \}$$

there are now 5 independent functions. therefore,

$$\boxed{\dim(V_2) = 5}$$

for a general N , we always have:

- one constant function 1,
- N cosine functions corresponding to $k = 1, \dots, N$,
- n sine functions corresponding to $K = 1, \dots, N$

so, the total number of basis functions is

$$1 + N + N$$

therefore,

$$\boxed{\dim(V_N) = 2N + 1}$$

1.2 Question 2

we have $S : V_N \rightarrow R_m$ defined by

$$S(f) = \begin{bmatrix} f(t_1) \\ f(t_2) \\ \vdots \\ f(t_m) \end{bmatrix}$$

(a)

we are given $m = 3$ with sampling points $t_1 = 0$, $t_2 = 0.5$, $t_3 = 1$ and

$$f(t) = 3 + \cos(2\pi t)$$

since S records the value of f at each sampling point,

$$S(f) = \begin{bmatrix} f(0) \\ f(0.5) \\ f(1) \end{bmatrix}$$

- at $t = 0$,

$$f(0) = 3 + \cos(2\pi(0))$$

$$f(0) = 3 + \cos(0)$$

since $\cos(0) = 1$,

$$f(0) = 4$$

- at $t = 0.5$,

$$f(0.5) = 3 + \cos(2\pi(0.5))$$

$$f(0.5) = 3 + \cos(\pi)$$

since $\cos(\pi) = -1$,

$$f(0.5) = 3 - 1 = 2$$

- at $t = 1$,

$$f(1) = 3 + \cos(2\pi(1))$$

$$f(1) = 3 + \cos(2\pi)$$

since $\cos(2\pi) = 1$,

$$f(1) = 3 + 1 = 4$$

putting all these together into the matrix, we get

$$S(f) = \begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix}$$

(b)

now we have

$$g(t) = \sin(12\pi t)$$

again,

$$S(g) = \begin{bmatrix} g(0) \\ g(0.5) \\ g(1) \end{bmatrix}$$

- at $t = 0$,

$$g(0) = \sin(12\pi(0)) = \sin(0) = 0$$

- at $t = 0.5$,

$$g(0.5) = \sin(12\pi(0.5)) = \sin(6\pi) = 0$$

- at $t = 1$,

$$g(1) = \sin(12\pi(1)) = \sin(12\pi) = 0$$

putting all these together into the matrix, we get

$$S(f) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(c)

for a function to be a linear map, we need

$$S(af + bg) = aS(f) + bS(g) \quad \text{for any functions } f, g \in V_N \text{ and scalars } a, b \in \mathbb{R}$$

starting with the LHS,

$$S(f) = \begin{bmatrix} (af + bg)(t_1) \\ (af + bg)(t_2) \\ \vdots \\ (af + bg)(t_m) \end{bmatrix}$$

we know that at each sample point t_i , the addition and scalar multiplication is

$$(af + bg)(t_i) = af(t_i) + bg(t_i)$$

therefore,

$$S(f) = \begin{bmatrix} af(t_1) + bg(t_1) \\ af(t_2) + bg(t_2) \\ \vdots \\ af(t_m) + bg(t_m) \end{bmatrix}$$

separating into two vectors:

$$S(f) = a \begin{bmatrix} f(t_1) \\ f(t_2) \\ \vdots \\ f(t_m) \end{bmatrix} + b \begin{bmatrix} g(t_1) \\ g(t_2) \\ \vdots \\ g(t_m) \end{bmatrix}$$

we can see that these two vectors are exactly $S(f)$ and $S(g)$ therefore,

$$S(af + bg) = aS(f) + bS(g)$$

hence,

$$\boxed{S \text{ is linear}}$$

(d)

(e)

(f)

(g)

(h)

2 Calculus

2.1 Question 3

a

b

c

2.2 Question 4

a

b

c